

ORIGINAL RESEARCH ARTICLE

Pathology of Genetic Variants in Spontaneous Coronary Artery Dissection

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BACKGROUND: Spontaneous coronary artery dissection (SCAD) is characterized by a separation of the coronary artery wall, causing myocardial infarction and sudden death. This study is one of the first to clarify the impact of pathological genetic variants in candidate SCAD-related genes on the histopathological and morphological properties of human SCAD lesions.

METHODS: A total of 28 SCAD cases were selected from the CVPPath Autopsy Registry. Histological differences in collagen and smooth muscle cells were compared among 3 groups: culprit SCAD (C-SCAD), defined as the coronary segment containing a dissection in cases with SCAD; nonculprit SCAD, defined as unaffected coronary segments (ie, healthy arteries) in cases with SCAD; and non-SCAD controls. Whole-exome sequencing was performed, and the identified variants were filtered to isolate pathogenic or likely pathogenic variants. Protein expression corresponding to the identified variants was assessed by immunostaining.

RESULTS: Collagen content was significantly reduced in C-SCAD lesions compared with controls, and immunohistochemical staining for smoothelin was reduced in the media of C-SCAD lesions compared with controls, whereas there were no significant differences between C-SCAD and nonculprit SCAD. Further, the nuclear height/width ratio of smooth muscle cells was increased, suggesting smooth muscle cell phenotypic modulation. In whole-exome sequencing analysis, pathogenic or likely pathogenic variants were detected in 25% of cases, which showed a higher frequency of multivessel dissection. In SCAD cases with pathogenic or likely pathogenic variants in *COL3A1* (collagen type III alpha 1), *FBN1* (fibrillin-1), or *FLNA* (filamin A), the expression of the corresponding protein was reduced in the media.

CONCLUSIONS: Reduced medial collagen and smooth muscle cell phenotypic modulation may be possible predisposing conditions for SCAD, regardless of the presence of pathogenic or likely pathogenic variants. The presence of pathogenic variants in extracellular matrix-related genes may further compromise arterial wall medial integrity, contributing to more severe disease. These findings provide novel pathological and genetic insights into the pathogenesis of SCAD and may inform future diagnostic and therapeutic strategies.

Key Words: collagen ■ coronary artery dissection, spontaneous ■ gene ■ pathology ■ smooth muscle cells

Spontaneous coronary artery dissection (SCAD) is characterized by a structural disruption or tear within the layers of the coronary artery wall, in which blood accumulating within the dissected layers compresses the lumen, resulting in myocardial ischemia/infarction.¹ SCAD accounts for 1% to 4% of acute

coronary syndromes and is implicated in 0.5% of sudden cardiac deaths.^{2–4} Typically, SCAD affects relatively young individuals, with a mean age of 44 to 53 years, and the majority of patients are women (87% to 95%).¹ It is believed that SCAD accounts for 35% of heart attacks in women <50 years of age, and its risk factors include

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Clinical Perspective

What Is New?

- Histological analysis of human lesions with spontaneous coronary artery dissection (SCAD) revealed reduced collagen content and smooth muscle cell phenotypic modulation in the coronary tunica media, providing new insights into their association with SCAD.
- The presence of pathogenic loss-of-function variants in extracellular matrix–related genes was seen in one-quarter of cases and was associated with an increased prevalence of multivessel dissection and reduced expression of the encoded proteins, suggesting the importance of genetic testing in subjects with a family history of SCAD.

What Are the Clinical Implications?

- Loss of medial collagen and smooth muscle cell phenotypic modulation was found in all cases of SCAD regardless of the presence of pathogenic/likely pathogenic variants, and clinical research should focus on early identification of these at-risk individuals.
- Genetic evaluation of decedents may be warranted in individuals with a family history of SCAD, particularly in those presenting with multivessel dissection or sudden cardiac death attributable to SCAD, to identify individuals who may be at increased risk of sudden cardiac death.

Nonstandard Abbreviations and Acronyms

COL3A1	collagen type III alpha 1
ECM	extracellular matrix
FBN1	fibrillin-1
FLNA	filamin A
PKD1	polycystic kidney disease 1
P/LP	pathogenic/likely pathogenic
SCAD	spontaneous coronary artery dissection
SMC	smooth muscle cell
SNP	single-nucleotide polymorphism
TGFB1	transforming growth factor beta 1

pregnancy, hypertension, extreme physical exertion, and acute emotional stress.⁵

Histopathologically, SCAD is defined by the presence of a hematoma formed either within the dissected media or between the media and the external elastic lamina, which compresses the arterial lumen. Communication between the lumen and dissected lumen is rare, suggesting that structural vulnerability within the coronary arterial media plays a critical role in the pathogenesis of SCAD. However, histopathological investigations of SCAD are limited to ≈100

case reports,^{3,6–9} and the pathophysiological mechanisms driving their formation remain insufficiently elucidated.

The extracellular matrix (ECM) is a critical determinant of vascular wall structural integrity and elasticity. Recent genomewide association studies of SCAD cases have revealed the involvement of single-nucleotide polymorphisms (SNPs) in ECM-related genes.^{10,11} Genetic variants associated with connective tissue disorders such as Ehlers-Danlos syndrome (*COL3A1* [collagen type III alpha 1]), Marfan syndrome (*FBN1* [fibrillin-1]), and Loeys-Dietz syndrome (*TGFB1* [transforming growth factor beta 1]) have also been reported in 1 in 5 individuals with SCAD.^{12–14} Given that SCAD does not demonstrate clear familial clustering across multiple generations and lacks features of a strictly monogenic disorder, it is considered a heterogeneous disease underpinned by multiple ECM-related genetic variants.¹⁵

Vascular smooth muscle cells (SMCs) are the predominant cell type comprising the coronary tunica media and play an essential role in maintaining vessel wall structural integrity by producing factors such as collagen. We hypothesized that genetic variants in genes associated with ECM production in VSMCs may be important in the pathogenesis of SCAD via ECM dysfunction.

In this study, we performed whole-exome sequencing on 20 SCAD cases from the CVPath Institute Sudden Death Autopsy Registry, correlating the presence of known pathogenic/likely pathogenic (P/LP) variants with histopathological and morphological findings from these cases, including the distribution of collagen matrix, and functional characteristics of SMCs in the coronary tunica media to provide new insights into the pathogenic mechanisms of SCAD. We believe this to be the first systematic study correlating genetics to vascular pathological abnormalities in cases of SCAD.

METHODS

Study Population

From 2004 to 2025, CVPath Institute was referred 944 autopsy hearts from adult subjects (≥18 years of age) dying from sudden coronary death from Maryland, Virginia, and Iowa in the United States and Nova Scotia in Canada. Sudden coronary death was defined as witnessed sudden unexpected death within 6 hours of the onset of symptoms or within 24 hours from being last seen in a stable medical condition.¹⁶ Ethical approval for this study was granted by the CVPath institutional review board. The requirement for informed consent was waived by the ethics committee as this study used only autopsy material. The data that support the findings of this study are available from the corresponding author upon reasonable request. Detailed methods for the study population are described in the [Supplemental Methods](#)

Histopathological Analysis

Detailed methods for histological and immunohistochemical analysis are described in the [Supplemental Methods](#).

Identification of P/LP Variants in SCAD-Related Genes

Whole-exome sequencing was performed at a single center at the University of Maryland. The methods of DNA isolation and identification of P/LP variants are described in the [Supplemental Methods](#).

Statistical Analysis

Continuous variables are presented as mean±SD or median (interquartile range), and categorical variables are expressed as percentages (N). The normality of distribution was assessed using the Shapiro-Wilk test. Continuous variables with a normal distribution were compared between groups using one-way analysis of variance (ANOVA), whereas non-normally distributed variables were compared using the unpaired Mann-Whitney *U* test. For comparisons between culprit SCAD (C-SCAD) and nonculprit lesions within the same SCAD patients (NC-SCAD), the Wilcoxon signed-rank test was applied. Categorical variables were compared using Pearson's χ^2 test. Non-normally distributed continuous variables were compared between the 3 groups (ie, control, C-SCAD, and NC-SCAD), using the Kruskal-Wallis test. Post hoc pairwise comparisons were conducted using Dunn's test, and statistical significance was defined as $P<0.05/3=0.0167$. Otherwise, a *P* value of <0.05 was considered statistically significant. For the genetic analysis, a *P* value threshold of 0.05 per number of variants was considered statistically significant. All statistical analyses were performed using JMP software version 16.1.0 (SAS Institute) or GraphPad Prism version 10.0.1 (GraphPad Software).

RESULTS

Patient and Lesion Characteristics of SCAD Cases From the CVPPath Autopsy Registry

[Figure S1](#) shows the study flow. Culprit lesion distribution and clinical characteristics from a total of 28 SCAD subjects from the CVPPath Sudden Death Registry were evaluated ([Table S1](#)). The median age of subjects with a diagnosis of SCAD was 43 years [interquartile range, 36–50], and 86% of these subjects were women. A history of fibromuscular dysplasia was present in 14% of cases, and pregnancy-associated SCAD accounted for 25% of cases. The majority of cases involved the left anterior descending artery (68%), whereas 29% involved the left main trunk. Arterial dissections were located at the proximal portion in 68% of anterior descending artery cases, 67% of left circumflex artery cases, and 36% of right coronary artery cases. Additionally, 25% of cases exhibited multivessel dissections. In contrast, previous clinical studies have reported proximal involvement in only 5% to 15% of cases and multivessel disease in $\approx 10\%$,¹⁷ suggesting that the present data set represents more severe forms of the disease. Of these, 97% showed dissection between the external elastic lamina border and outer media, whereas only 3% involved the inner media ([Figure 1A and 1B](#)). In 78% of lesions, the dissection ex-

tended over an arc of >271 degrees around the vessel circumference ([Figure 1C](#)). These findings suggest that extensive dissection may represent a significant risk factor for sudden death in SCAD.⁷

Comparison of Pathological Findings Between SCAD and Control Arteries

Additional histopathological characterizations were performed on a subset of SCAD cases for which formalin-fixed paraffin-embedded blocks were available ($n=20$). Twenty age- and sex-matched noncardiac death cases were selected from the CVPPath Autopsy Registry for comparative analysis (ie, control group). There were no significant differences in patient background or clinical characteristics between the SCAD and control groups ([Table S2](#)). SCAD is recognized as a nonatherosclerotic disease and has been associated with a reduced risk of atherosclerosis.¹⁸ Therefore, we compared the atherosclerotic grade in sections from culprit SCAD lesions (C-SCAD; $n=178$ sections from 20 cases), nonculprit lesions within the same SCAD patients (NC-SCAD; $n=95$ sections from 11 cases), and control arteries ($n=184$ sections from 20 cases). Lesions equal to or more advanced than pathological intimal thickening were observed in 34% of control sections, whereas such lesions were identified in only 4% and 13% of C-SCAD and NC-SCAD sections, respectively ([Figure S2](#)).

Comparison of Medial Collagen Content Between SCAD and Control Arteries

To compare arterial collagen content between SCAD cases and controls, we evaluated coronary artery sections stained with Masson's trichrome and Picrosirius Red (2 histological stains used to differentiate collagen from other tissues) and quantified staining density within the media. We also compared collagen content between culprit lesions and nonculprit lesions within the same subjects for which both sections were available. To eliminate the influence of atherosclerosis, only sections with lesions less advanced than pathological intimal thickening were included in the quantitative assessments. We found that the percentage of collagen content in the media was significantly lower in C-SCAD compared with control (15.8% [12.9–22.9] versus 36.5% [31.0–48.9]; $P<0.001$ in Masson's trichrome assessment; and 17.9% [15.1–24.8] versus 40.8% [35.3–48.8]; $P<0.001$ in Picrosirius Red assessment). However, no significant differences in collagen content by either Masson's or Picrosirius Red staining were observed between C-SCAD and NC-SCAD within the same individuals (14.5% [11.9–23.7] versus 17.7% [16.7–21.8]; $P=0.58$ in Masson's trichrome assessment; and 17.0% [14.8–24.1] versus 22.9% [16.7–26.5]; $P=0.52$ in Picrosirius

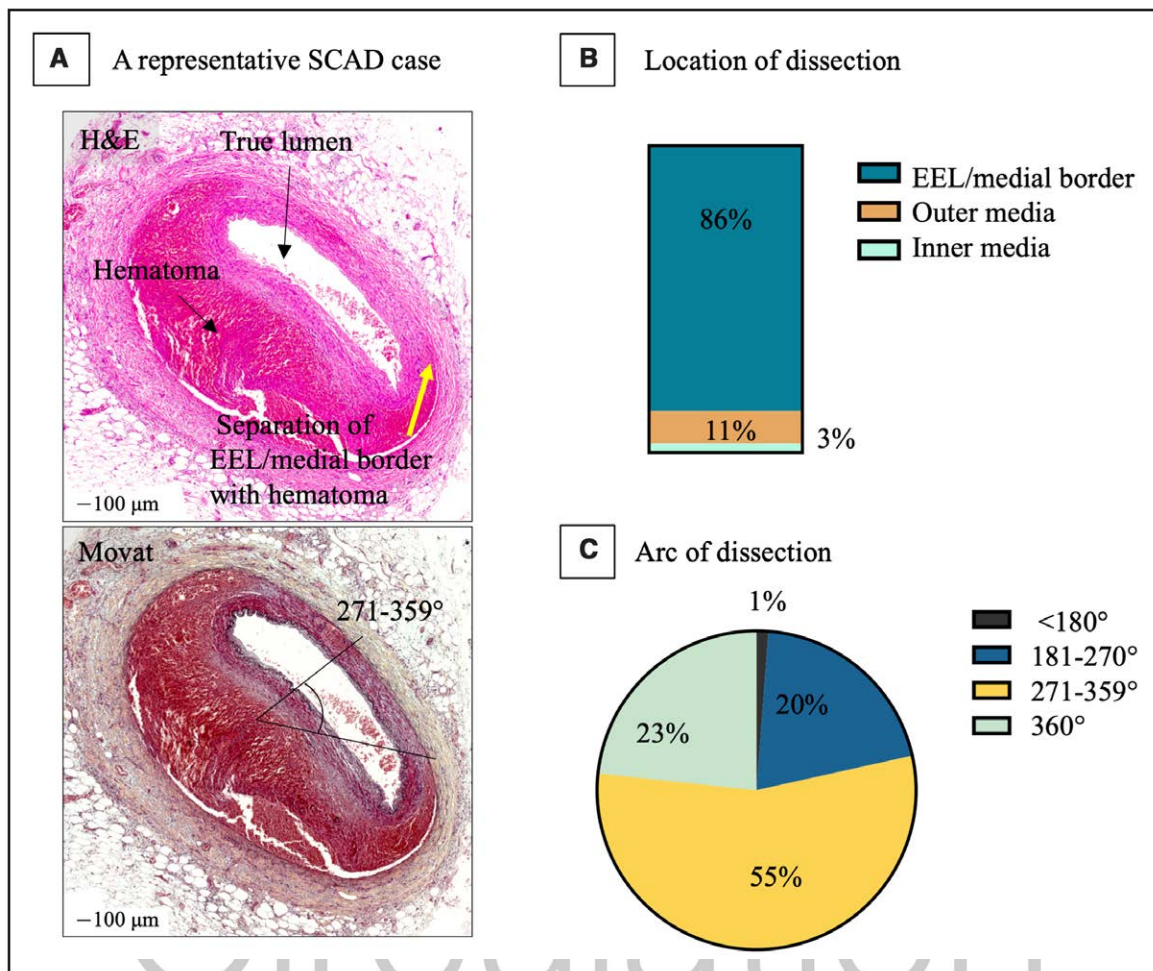


Figure 1. Location of dissection and dissection arc in human SCAD lesions.

A, Representative cross-section stained with H&E (top) and Movat Pentachrome (bottom) of posterolateral artery from a 32-year-old 2-week post-partum woman illustrating the location and arc of dissection in the subject of SCAD. **B**, Bar graph showing the distribution of the location of dissection, including the EEL border, outer media, and inner media. **C**, Pie chart shows the distribution of arc of dissection, including <180, 181–270, 271–359, and 360 degrees. A total of 28 cases from culprit lesions in subjects with SCAD were used for calculations in **B** and **C**. EEL indicates external elastic lamina; H&E, hematoxylin and eosin stain; and SCAD, spontaneous coronary artery dissection.

Red assessment; Figure 2I). This suggests that loss of medial collagen throughout the coronary tree may be associated with SCAD rather than a process limited to the culprit SCAD lesion.

Evaluation of Medial SMC Phenotype

We next investigated the phenotypic state of SMCs in the media via immunostaining to examine whether changes in medial SMC might be associated with SCAD. Alpha smooth muscle actin (α SMA) was used as a general SMC marker. The percentage of α SMA-positive area in the media did not show differences between C-SCAD and control lesions (47.9% [40.8–52.0] versus 45.8% [41.7–51.6]; $P=0.95$), or between C-SCAD and NC-SCAD (47.8% [40.9–52.2] versus 47.0% [34.8–54.0]; $P=0.58$). In contrast, expression of smoothelin, a contractile SMC-specific marker, was significantly lower in C-SCAD media compared with control media (8.8%

[3.8–20.4] versus 16.4% [8.5–30.0]; $P<0.05$), although no significant differences were observed between C-SCAD and NC-SCAD (17.2% [3.1–23.7] versus 12.0% [6.9–14.8]; $P=0.46$; Figure 2I). Smoothelin is known to disappear early during the dedifferentiation of vascular SMCs. On the other hand, α SMA is used as a general marker for the presence of SMCs because it is consistently retained even in synthetic SMCs.¹⁹ This suggests that although the overall abundance of SMCs is unchanged, the proportion of contractile SMCs is reduced in SCAD, possibly reflecting phenotypic modulation. Of note, given that specific immunohistochemical markers to indicate SMC modulation are limited, assessment of SMC morphological characteristics was used to further interrogate potential SMC phenotype. In particular, we evaluated SMC morphology by quantification of the nuclear height-to-width (HW) ratio. Contractile SMCs typically show spindle-shaped morphology but undergo a transition to a rhomboid shape during phenotypic

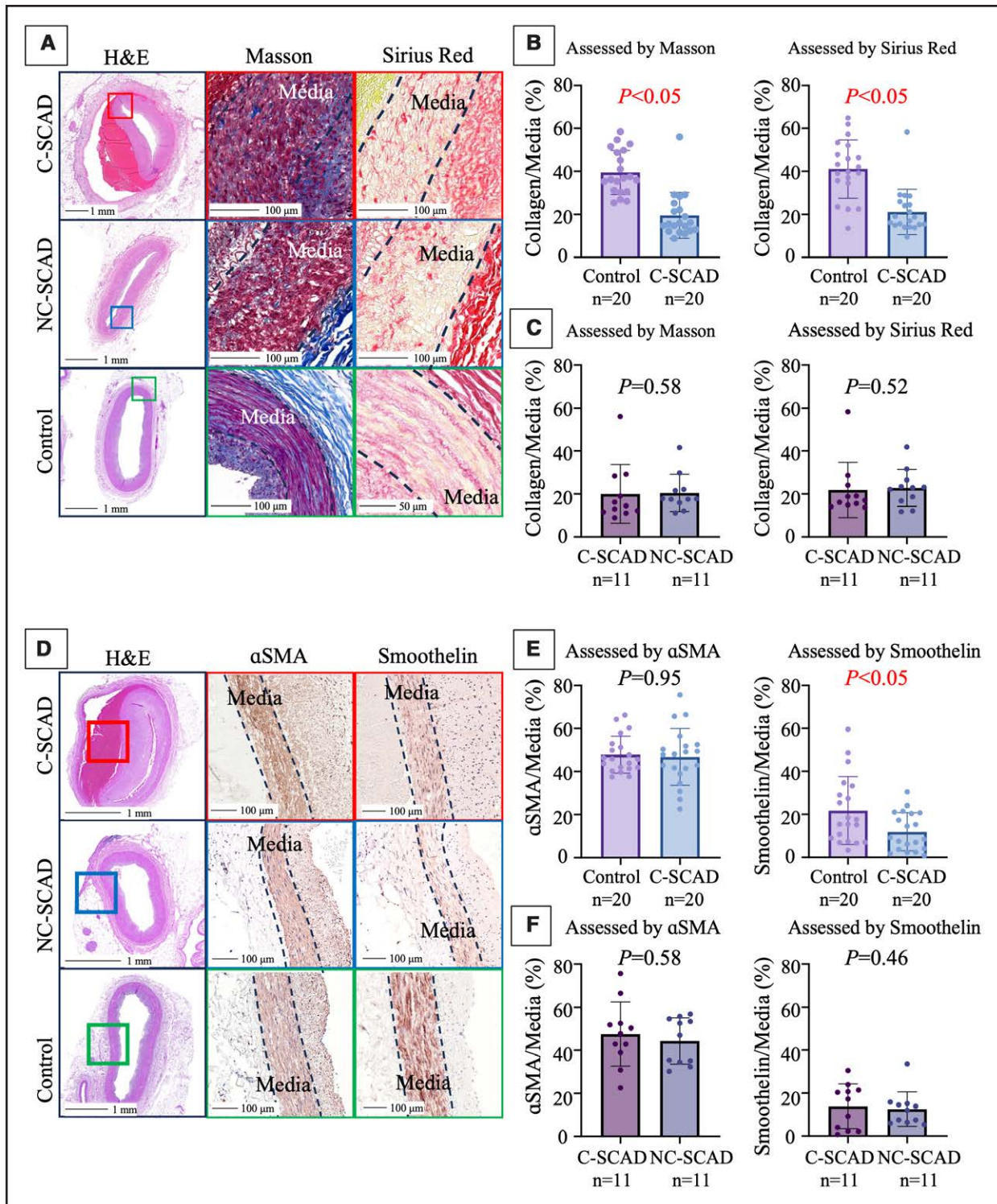


Figure 2. Evaluation of collagen content and SMCs in the coronary tunica media.

A, Representative images of coronary artery sections in SCAD and control cases. Low-power images are illustrated with H&E staining. Images from Masson's trichrome staining and Picrosirius Red staining are shown with high-power images of the boxed area in the H&E staining images. **B**, Scatter plots with bar graphs comparing control (n=20) and C-SCAD (n=20), assessed by Masson and Picrosirius Red staining. **C**, Scatter plots with bar graphs comparing C-SCAD (n=11) and NC-SCAD (n=11), assessed by Masson and Picrosirius Red staining. **D**, Representative images of the coronary artery in SCAD and control cases. Low-power images are illustrated with H&E staining. The images with α SMA and smoothelin immunostaining are shown with high-power images of the boxed area in each H&E staining image. **E**, Scatter plots with bar graphs comparing control (n=20) vs C-SCAD (n=20), assessed by α SMA and smoothelin. **F**, Scatter plots with bar graphs comparing C-SCAD (n=11) vs NC-SCAD (n=11), assessed by α SMA and smoothelin. Unpaired Mann-Whitney *U* test was applied for **B** and **E**. Wilcoxon signed-rank test was applied for **C** and **F**. *P* values that reached statistical significance are shown in red. α SMA indicates alpha smooth muscle actin; C-SCAD, culprit spontaneous coronary artery dissection; H&E, hematoxylin and eosin stain; and NC-SCAD, nonculprit SCAD.

switching.^{20,21} The nuclear HW ratio was significantly lower in the control than C-SCAD and NC-SCAD (19 and 11 cases, respectively, control [18 cases]: 13.8% [11.3–19.3]; n=360 cells versus C-SCAD: 25.5% [19.8–32.7]; n=380 cells and NC-SCAD: 24.0% [17.8–32.2]; n=220 cells; $P<0.0001$, respectively), whereas C-SCAD and NC-SCAD were comparable ($P=0.79$; Figure 3), further suggesting phenotypic modulation from contractile SMCs. Collectively, our findings suggest that reduced collagen content and SMC phenotypic modulation are associated with SCAD pathophysiology.

Identification of P/LP Genetic Variants in SCAD-Related Genes

After our comprehensive histopathological analysis, we conducted whole-exome sequencing on the 20 SCAD lesions in our autopsy database for which formalin-fixed paraffin-embedded blocks were available. All exonic variants passing quality thresholds as described in the Materials and Methods were filtered to identify those within genes previously implicated in SCAD pathogenesis or associated with the regulation of the ECM (Table S3). From all high-confidence variants in the exome, 2601 were detected within SCAD-related genes.^{10,22–24} To fo-

cus our analysis on those variants that were most likely to underlie disease development, we selected those that were classified as P/LP based on Franklin genomic platform annotations (Figure S3). Within SCAD-related candidate genes, we ultimately identified 6 P/LP variants (rs1553292145, rs1553507614, rs587779541, rs193922243, rs1555450968, and rs2067761804) in *TGFB2*, *COL3A1*, *FBN1*, *PKD1* (polycystic kidney disease 1), and *FLNA* (filamin A) within 5 of 20 cases. Each of these variants has been strongly implicated in diseases known to cause arterial dissection. Tables S4 and S5 show detailed information derived from the Franklin genomic platform for each DNA variant. Notably, multi-vessel dissections were significantly more frequent in SCAD cases with P/LP variants compared with cases without P/LP variants (60% versus 13%; $P<0.05$; Figure S4; Table).

Protein-Level Impacts of P/LP Variants Within Human SCAD Lesions

We next evaluated the effect of select P/LP variants on protein expression and fibrotic content within the coronary arterial media via histological and immunohistochemical staining. In particular, we focused on 4 cases

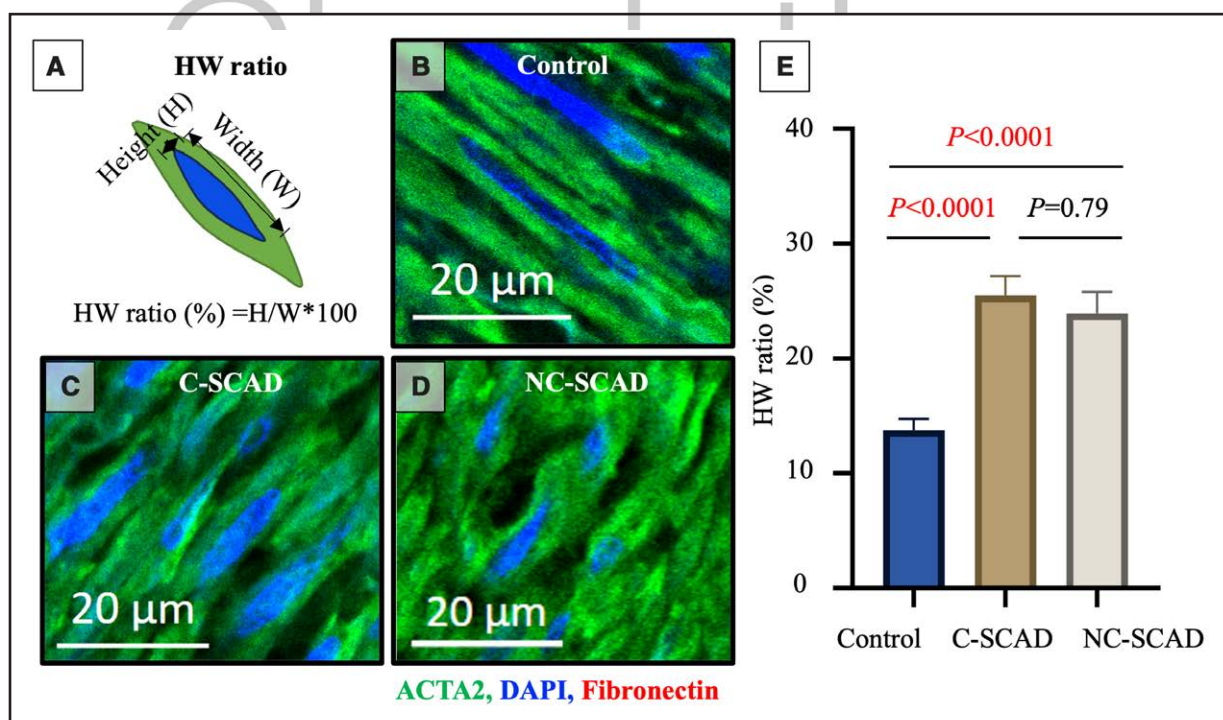


Figure 3. Evaluation of nuclear shape in medial SMCs in SCAD lesions.

A, Nuclear HW ratio was calculated as the value of nuclear height divided by the nuclear width of VSMCs in the media. **B**, **C**, and **D**, Representative confocal images of VSMCs in media in control, C-SCAD, and NC-SCAD media, respectively. **E**, Bar graphs of quantitative analysis for the HW ratio comparing control (n=360 cells), C-SCAD (n=380 cells), and NC-SCAD (n=220 cells). The HW ratio was significantly lower in control than in C-SCAD and NC-SCAD; however, there was no significant difference between C-SCAD and NC-SCAD. Kruskal-Wallis test, followed by Dunn's post hoc test, was applied for **E**. P values that reached statistical significance are shown in red. C-SCAD indicates culprit spontaneous coronary artery dissection; DAPI, 4',6-diamidino-2-phenylindole; H&E, hematoxylin and eosin stain; HW, height-to-width; NC-SCAD, nonculprit SCAD; SMC, smooth muscle cell; and VSMC, vascular smooth muscle cell.

Table. P/LP Variants Identified by WES in Each SCAD Case From CVPPath Autopsy Registry

Case no.	Age (y)	Sex	Race	Lesion	P/LP variants
1	42	F	White	SVD	
2	47	F	White	SVD	
3	58	M	White	SVD	
4	34	F	Black	MVD	
5	36	F	White	SVD	
6	41	F	White	SVD	
7	48	M	Hispanic	SVD	
8	54	M	White	SVD	
9	40	F	Black	SVD	
10	39	M	White	SVD	
11	36	F	White	SVD	
12	38	F	Black	SVD	
13	43	F	White	SVD	
14	32	F	Black	MVD	
15	61	F	Black	SVD	
16	47	F	White	SVD	<i>PKD1</i> (rs1555450968)
17	42	F	White	SVD	<i>FLNA</i> (rs2067761804)
18	42	F	White	MVD	<i>COL3A1</i> (rs1553507614)
19	41	F	N/A	MVD	<i>COL3A1</i> (rs587779541)
20	62	F	White	MVD	<i>FBN1</i> (rs193922243) <i>TGFB2</i> (rs1553292145)

COL3A1 indicates collagen type III alpha 1; *FBN1*, fibrillin-1; MVD, multivessel dissection; N/A, not applicable; *PKD1*, polycystic kidney disease 1; P/LP, pathogenic/likely pathogenic; SVD, single-vessel dissection; *TGFB2*, transforming growth factor beta 2; and WES, whole-exome sequencing.

that harbored P/LP variants in genes encoding key structural proteins comprising the cellular cytoskeleton or the ECM (case No. 17, c.1087C>T in *FLNA*; case No. 18, c.953G>A in *COL3A1* chain; case No. 19, c.2285G>A in *COL3A1*; and case No. 20, c.8226+5G>A in *FBN1*; Table). Age- and artery-matched control cases were selected from the control group for comparison staining. To obtain detailed information about collagen composition, polarized light microscopy of Picrosirius Red-stained sections was used for visual comparison. In these Picrosirius Red images, nonfragmented collagen, such as type I collagen, appeared yellow to orange-red, whereas fragmented collagen, such as type III collagen, appeared green.²⁵ In all 4 cases, birefringent collagen fibers, both red and green, were visually reduced compared with control cases, suggesting overall loss of collagen (Figures 4 and 5). Immunohistochemical analysis of type III collagen similarly revealed substantially reduced expression in the cases with *COL3A1* P/LP variants (cases 18 to 19) relative to controls (Figure 4). *FBN1*, encoded by *FBN1*, is a key ECM component that provides a scaffold for microfibrils, which are essential for the formation and maintenance of elastic fibers in the vascular system. The expression of *FBN1* was markedly lower in the culprit lesion of Case 20 that harbored a genetic variant in *FBN1* than in the control (Figure 5A). Filamin A, encoded by

FLNA, is an actin-binding protein essential for forming a network of intracellular protein filaments. Similarly, the percentage of filamin A in the media was lower in the culprit lesion of case 17 with a variant in *FLNA* compared with the control (Figure 5B). These findings suggest that P/LP variants are associated with reduced expression of their encoded proteins in the arterial media, which may contribute to the development of multivessel dissection.

DISCUSSION

In this study, we histologically evaluated collagen content and SMCs in the coronary tunica media and examined DNA variants and their associated protein expression in subjects dying suddenly of SCAD from the CVPPath Sudden Death Registry. This study is the first systematic study to correlate genetics to vascular pathological abnormalities in cases of SCAD. We report 2 novel findings. First, reduced collagen content and SMC phenotypic modulation were observed in the coronary tunica media in subjects with SCAD versus control arteries. Second, pathogenic variants were associated with a higher prevalence of multivessel dissection, and expression of the protein encoded by pathogenic variants was reduced in the coronary tunica media in subjects with SCAD carrying P/LP variants for these proteins. These integrated

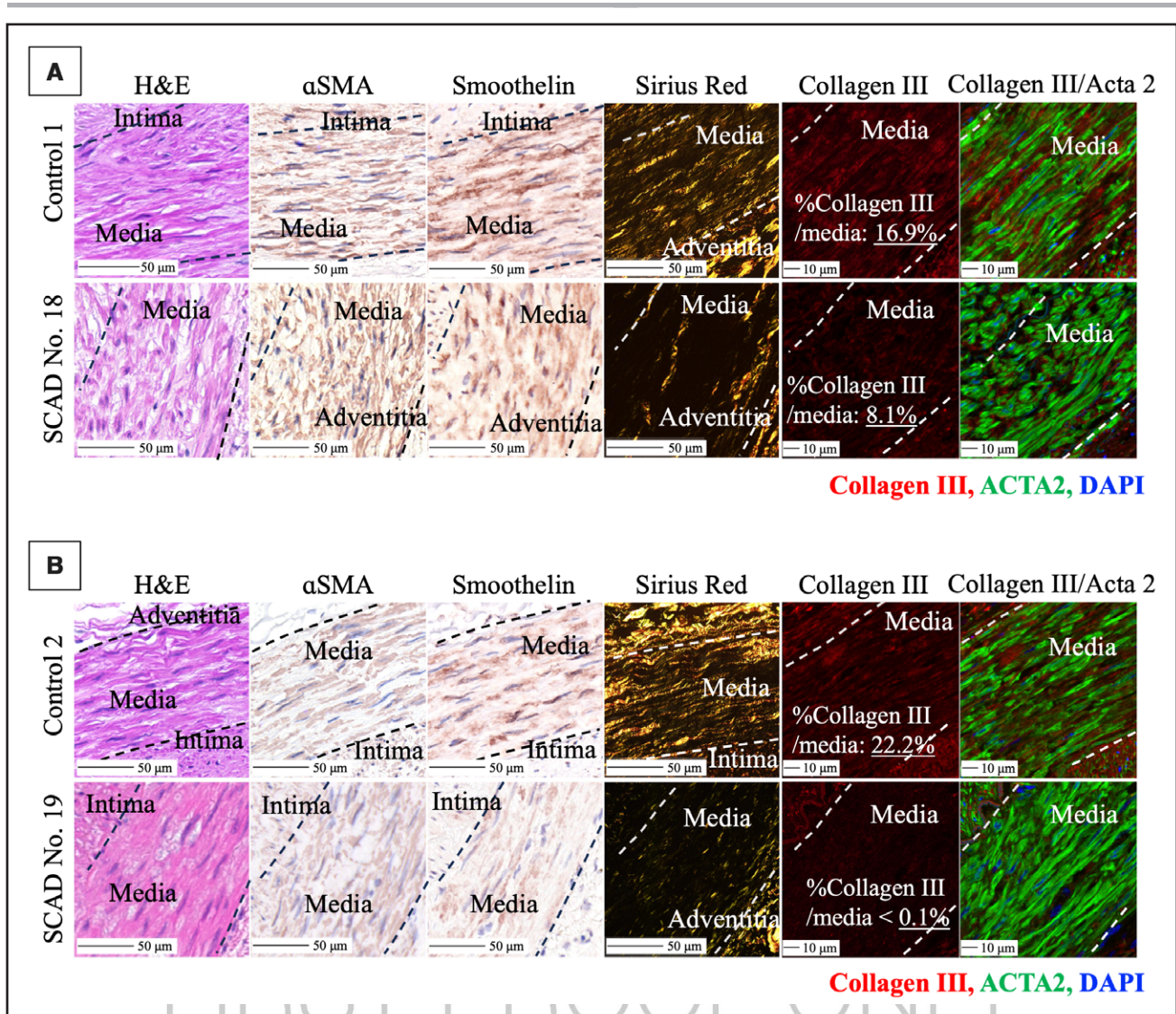


Figure 4. Immunohistochemical analysis of SCAD lesions with P/LP COL3A1 (collagen type III alpha 1) variants.

A, The control sample is from a 39-year-old woman matched for age, sex, and coronary artery location with case No. 18 in SCAD. The culprit lesion is from the LAD in a 42-year-old woman with SCAD carrying a *COL3A1* variant (case No. 18 in the Table). **B**, The control sample is from a 39-year-old woman matched for age, sex, and coronary artery location with case No. 19 in SCAD. The culprit lesion shows the LCX from a 41-year-old woman with SCAD carrying a *COL3A1* variant (case No. 19 in the Table). High-power images are illustrated with H&E, immunohistochemistry (IHC) for αSMA and smoothelin, the polarized light images of Picrosirius Red staining, and immunofluorescence for collagen (collagen III single image and collagen III/ACTA2/DAPI merged image). Collagen III evaluated by immunofluorescence was less in SCAD (8.1% [6.3–10.3] and 0.02% [0.02–0.05], respectively) compared with control (16.9% [15.8–19.2] and 22.2% [18.3–23.8], respectively). The percentage of collagen III/media was calculated from 5 and 6 control sections and 4 and 5 SCAD sections, respectively. Unpaired Mann-Whitney *U* test was applied. αSMA indicates alpha smooth muscle actin; ACTA2, actin alpha2, smooth muscle; DAPI, 4',6-diamidino-2-phenylindole; H&E, hematoxylin and eosin stain; LAD, left anterior descending artery; LCX, left circumflex artery; P/LP, pathogenic/likely pathogenic; and SCAD, spontaneous coronary artery dissection.

results using both histological and genetic findings may collectively further our understanding of the development and risk for progression of SCAD.

SCAD is a condition characterized by spontaneous structural disruption within the medial layer of the coronary arterial wall. Since its first report in 1931,²⁶ some studies have investigated its pathophysiology.^{6–9} However, because of the rarity of cases, data, particularly regarding pathological findings, remain limited. The current study is the first pathological analysis of SCAD

to focus on collagen content and SMCs in the coronary tunica media, demonstrating reduced collagen content and altered SMC phenotypic modulation. The coronary tunica media predominantly consists of ECM, including collagen content, and SMCs. SMCs are broadly classified into contractile and synthetic types. The primary function of contractile SMCs is to regulate vascular tone in healthy arteries.²⁷ On the other hand, synthetic SMCs are characterized by increased migratory capacity and enhanced secretion of proteins such as components of

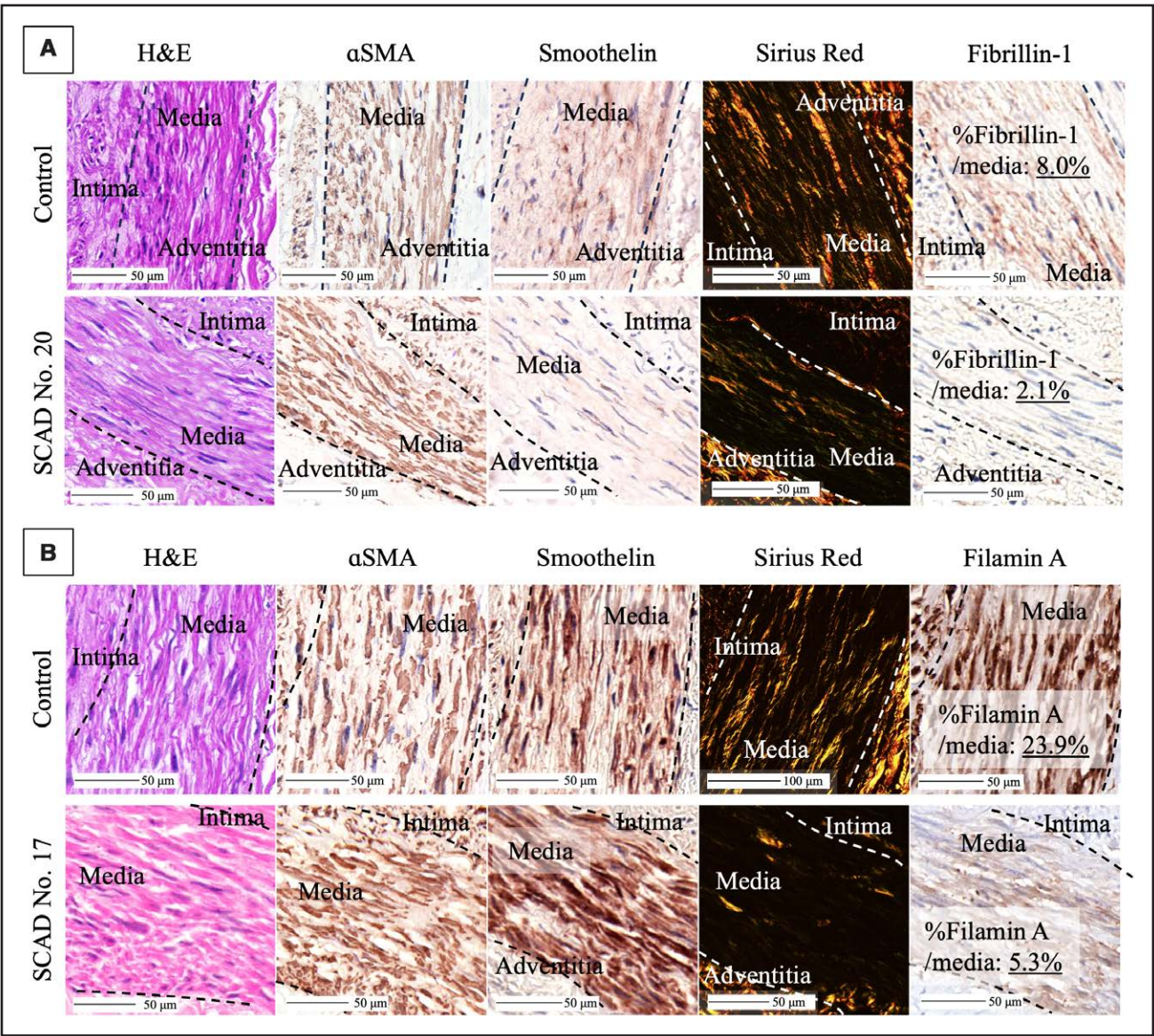


Figure 5. Representative SCAD case with each *FBN1/TGFB2* and *FLNA* variants compared with the control case.
A, The control case shows the LCX from a 61-year-old control woman. The SCAD case shows culprit lesions from the LCX from a 62-year-old woman with SCAD carrying an *FBN1/TGFB2* variant (case No. 20 in the Table). The percentage of FBN1/media was less in C-SCAD (2.1% [0.6–6.1]) compared with control (8.0% [7.2–8.4]). The percentage of FBN1/media was calculated from 5 control sections and 4 C-SCAD sections. **B**, The control sample shows LCX from a 44-year-old woman matched for age, sex, and coronary artery location with case No. 17 in SCAD. SCAD lesion shows the cross-sections from the LCX of a 42-year-old woman with SCAD carrying an *FLNA* variant (case No. 17 in the Table). The percentage of filamin A/media was less in SCAD (5.3% [1.6–12.1]) compared with control (23.9% [22.4–25.8]). The percentage of filamin A/media was calculated from 4 control sections and 4 SCAD sections. Unpaired Mann-Whitney *U* test was applied. High-power images are illustrated with H&E, immunohistochemistry (IHC) for αSMA and smoothelin, the polarized light images of Picrosirius Red staining, and IHC for FBN1 or filamin-A, respectively. αSMA indicates alpha smooth muscle actin; C-SCAD, culprit spontaneous coronary artery dissection; FBN1, fibrillin-1; H&E, hematoxylin and eosin stain; LCX, left circumflex artery; SCAD, spontaneous coronary artery dissection; and TGFB1, transforming growth factor beta 1.

the ECM, which contribute to vascular remodeling and the development of atherosclerosis.²¹ The structural integrity of arteries is maintained by factors such as the biological properties of the ECM, regulation of vascular tone, and tissue coagulation mechanisms. Nonetheless, it has been reported that synthetic SMCs encompass various subtypes, including fibroblast-like, adipocyte-like, macrophage-like, T cell-like, and mesenchymal-like

SMCs.²⁸ The proteins synthesized by synthetic SMCs vary depending on their subtype, and this phenotypic modulation is important for proper vascular function, as arterial integrity is impacted by ECM biology, especially by the collagen family of proteins and matrix metalloproteases (MMPs), which cause the degradation of collagen in the ECM.^{21,29} Several reports indicate that collagen degradation attributable to MMPs and SMC phenotypic

modulation into synthetic and mesenchymal-like SMCs are related to the development of aortic aneurysms and aortic dissection.^{30,31} Although it remains unclear whether smooth muscle cells adopt a phenotypic modulation as a consequence of collagen loss or whether abnormal phenotypic modulation itself leads to reduced collagen deposition in SCAD, our findings suggest that collagen loss could be an underlying contributor to SCAD. Because analyses were restricted to sections with minimal atherosclerosis, differences in atherosclerotic burden between groups raise the possibility that residual variations in vessel composition may have influenced collagen comparisons.

Over the past decade, significant progress has been made in understanding the genetic basis of SCAD, with variants within ECM-related genes such as *COL1A1*, *FBN1*, *TGFBR1*, *ADAMTSL4*, *EDN1*, and *LRP1* being reported in 5% to 8% of patients,^{11,13,32–34} suggesting that ECM abnormalities may contribute to its pathogenesis. We performed whole-exome sequencing in our SCAD cohort. Despite the consistent finding of coronary medial collagen loss in SCAD cases, we did not identify any P/LP DNA variants that were consistently associated with collagen loss across the entire cohort. Meanwhile, among the 20 cases analyzed for DNA variants in the current study, 5 cases (25% of the total) carried P/LP variants.

To support the validity of the whole-exome sequencing findings, we further reviewed the available medical and family history of the 5 subjects with identified P/LP variants. The subject with the *FLNA* variant (rs2067761804) had a history of mitral valve prolapse, which has been reported to be associated with *FLNA* variants.³⁵ For the subject with the *COL3A1* variant (rs1553507614), this individual had a family history notable for maternal death resulting from a brain aneurysm at 42 years of age. Another case with a *COL3A1* variant (rs587779541) had previously undergone genetic testing and was diagnosed as a carrier of *COL3A1* variant. The subject harboring both *FBN1* (rs193922243) and *TGFBR2* (rs1553292145) variants had a tall stature (180 cm), which is one of the features of Marfan syndrome, but no history of obvious aortic aneurysm was noted in the autopsy report. For the subject with *PKD1* (rs1555450968), there was not enough information provided about medical history.

Our findings indicate a notably higher frequency of P/LP variants than previously reported in clinical studies, which may be attributable to that fact that the cohort study consisted of subjects with sudden cardiac death directly attributable to SCAD. Others have reported similar findings in higher risk cohorts of SCAD subjects.¹² Furthermore, cases with P/LP variants were more likely to present with multivessel dissection compared with those without P/LP variants; however, these findings should be interpreted cautiously because of the limited

sample size. Furthermore, the subset of cases selected for sequencing may not be representative of results from larger study populations in which subjects with SCAD survive the initial event and thus may limit the generalizability of the results.

In cases with evidence of P/LP variants, we validated the effect of these DNA variants by investigating the impact of the protein encoded by pathogenic variants on the coronary tunica media using immunohistochemistry. *COL3A1*-related disorders are most notably associated with Ehlers-Danlos syndrome. *COL3A1* variants in this cohort (rs1553507614 and rs587779541) involve a heterozygous G to A transition at the *COL3A1* gene. The mutation alters the amino acid sequence, preventing proper triple-helix formation and leading to structural weakness in collagen fibers.^{36,37} Type III collagen is a fibrillar collagen composed of 3 $\alpha 1(\text{III})$ chains. In adults, type III collagen is essential for the proper formation of type I collagen fibers in the cardiovascular system.³⁸ This may explain why both type I and type III collagen were reduced in the 2 cases carrying *COL3A1* variants.

FBN1, encoded by the *FBN1* gene, serves as a scaffold for microfibrils, essential for the formation and maintenance of elastic fibers in the vascular system. The c.8226+5G>A variant (rs193922243) leads to the skipping of exon 65, causing a frameshift. This mutant transcript escapes nonsense-mediated decay,³⁹ resulting in translation and accumulation of abnormal protein without degrading. Previous human data showed significantly lower *FBN1* expression in the culprit aortic dissection lesion in patients with Marfan syndrome compared with controls.⁴⁰ Our data also demonstrated a significant reduction in *FBN1* within the medial layer.

The *FLNA* gene encodes filamin A, a widely expressed actin-binding protein. Loss of function mutations in *FLNA* are known as the most common cause of periventricular nodular heterotopia, a disorder involving abnormal neuronal migration, and X-linked cardiac valvular dystrophy, both of which involve various abnormalities in soft connective tissues.⁴¹ Recently, *FLNA* has also been reported as a novel SCAD gene.⁴² Previous studies showed that *FLNA* knockdown reduced SMC migration⁴³ and led to higher mRNA and protein expression levels of MMP-9.⁴⁴ Taken together, these findings strongly suggest that abnormalities in SMCs and an ECM imbalance, driven by the influences of P/LP variants, may underlie the pathogenesis of SCAD.

The mechanisms underlying intramural hematoma formation within the arterial media in SCAD remain incompletely understood. In this study, through integrated pathological and genetic analyses, we demonstrated that abnormal SMC phenotypes and imbalanced ECM composition may contribute to the pathogenesis of SCAD. Additionally, the presence of P/LP DNA variants that may further disrupt the ECM homeostasis can predispose

affected arteries to more severe multi-arterial dissection. Given these findings, genetic examination may be warranted in individuals with a family history of SCAD, especially in cases involving multiple vessel dissections or sudden cardiac death attributable to SCAD, to identify those who may be at high risk for SCAD. Further work is needed to understand the polygenic basis for SCAD, which likely plays a role in the majority of cases.

Study Limitations

This study has several limitations. First, although it represents one of the largest pathological data sets of SCAD to date, the overall number of cases remains relatively small. Analyses incorporating larger cohorts are needed. In addition, because the cases presented here were derived from sudden death events, a detailed medical and family history were sometimes unavailable. Regarding the racial identity of the subjects studied here, Black individuals comprised 25% of our cohort. Previous clinical studies of SCAD have reported lower rates of Black individuals, in the range of 1% to 7%.^{1,45} The higher prevalence of Black individuals in this study should be interpreted cautiously, and larger studies, including both autopsy-based and clinically diagnosed SCAD cohorts, are needed to better understand the role of racial differences in fatal SCAD. Second, characterization of SMC phenotype by immunohistochemistry is technically challenging. Future work using RNA-based approaches such as spatial transcriptomics, alongside in vitro and animal studies, will be necessary to define SMC phenotypes and clarify these relationships. Third, the cross-sectional autopsy design of this study limits causal inference. The findings of loss of medial collagen throughout the coronary tree in SCAD cases should not be interpreted as definitive evidence of a systemic process, as the effect of postmortem changes cannot be excluded. Fourth, variant classification is an ongoing process that is continuously refined with the accumulation of new evidence. Finally, our cohort is comprised of sudden death autopsies, likely representing the most severe cases, and thus, comparisons with data generated from contemporary clinical SCAD populations that survive the initial event may be limited. Further genetic analyses and expanded collection of pathological data could add additional important information.

Conclusions

In SCAD cases resulting in sudden death, this study revealed reduced medial collagen content and abnormal SMC phenotypic modulation in the coronary media regardless of the presence of P/LP variants, with particularly severe arterial dissection observed in cases harboring P/LP variants. These results suggest that abnormalities in ECM structural and regulatory components

are associated with SCAD. Collectively, these findings highlight the need for genetic evaluation, particularly in patients with a family history, multivessel dissection, or SCAD-related sudden cardiac death, to identify individuals at increased risk, but also to better understand the causal genetic mechanisms underlying SCAD in cases without P/LP variants in which similar loss of collagen is seen.

ARTICLE INFORMATION

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Supplemental Material

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